

Bahan Ajar Pert 13

BAHAN AJAR - PERTEMUAN 13

Python: Perulangan FOR, WHILE + Studi Kasus

TP	BK 1.4, BK 2.1
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A. FOR DI PYTHON — range()

FOR di Python menggunakan `range()` .

range(stop) — dari 0 sampai stop-1

```
for i in range(5):  
    print(i)  
# Output: 0 1 2 3 4
```

range(start, stop) — dari start sampai stop-1

```
for i in range(1, 6):  
    print(i)  
# Output: 1 2 3 4 5
```

range(start, stop, step) — dengan langkah

```
for i in range(2, 11, 2):  
    print(i)  
# Output: 2 4 6 8 10  
  
for i in range(10, 0, -1):  
    print(i)  
# Output: 10 9 8 ... 1  
  
for i in range(3, 31, 3):  
    print(i)  
# Output: 3 6 9 ... 30
```

Translasi Pseudocode → Python (FOR)

Pseudocode	Python
FOR i ← 1 TO 5	for i in range(1, 6):
FOR i ← 0 TO 4	for i in range(5):
FOR i ← 2 TO 10 STEP 2	for i in range(2, 11, 2):
FOR i ← 10 DOWNT0 1	for i in range(10, 0, -1):
ENDFOR	(indentasi)

Contoh Program FOR

Cetak 1-10:

```
for i in range(1, 11):  
    print(i)
```

Tabel perkalian 7:

```
for i in range(1, 11):  
    print(f"7 × {i} = {7 * i}")
```

Jumlah 1+2+...+n:

```
n = int(input("Masukkan n: "))  
total = 0  
for i in range(1, n + 1):  
    total += i  
print(f"Total = {total}")
```

Deret genap 2-20:

```
for i in range(2, 21, 2):  
    print(i, end=" ")  
# Output: 2 4 6 8 10 12 14 16 18 20
```

B. WHILE DI PYTHON

WHILE mengulang **selama** kondisi TRUE.

Sintaks

```
while kondisi:  
    # kode diulang
```

Contoh 1: Cetak 1-5

```
i = 1  
while i <= 5:  
    print(i)  
    i += 1      # i = i + 1 - INGAT update!
```

Contoh 2: Input sampai 0

```
total = 0  
x = int(input("Angka (0=stop): "))  
while x != 0:  
    total += x  
    x = int(input("Angka (0=stop): "))  
print(f"Total = {total}")
```

Contoh 3: Tebak angka

```
rahasia = 7
tebakan = int(input("Tebak angka: "))
while tebakn != rahasia:
    if tebakn < rahasia:
        print("Terlalu kecil!")
    else:
        print("Terlalu besar!")
    tebakn = int(input("Coba lagi: "))
print("Selamat! Benar!")
```

Contoh 4: Faktorial dengan WHILE

```
n = int(input("n: "))
hasil = 1
i = 1
while i <= n:
    hasil *= i
    i += 1
print(f"{n}! = {hasil}")
```

Translasi Pseudocode → Python (WHILE)

Pseudocode	Python
WHILE x<10	while x < 10:
x ← x + 1	x += 1
ENDWHILE	(indentasi)

C. BREAK & CONTINUE

break — Keluar dari perulangan

```
# Berhenti saat pengguna mengetik 0
while True:
    x = int(input("Angka (0=stop): "))
    if x == 0:
        print("Bye!")
        break
    print(f"Kamu mengetik {x}")
```

continue — Loncat ke iterasi berikutnya

```
# Cetak angka ganjil dari 1-10 (skip genap)
for i in range(1, 11):
    if i % 2 == 0:
        continue
    print(i, end=" ")
# Output: 1 3 5 7 9
```

Perbandingan break vs continue

Perintah	Efek
break	Keluar dari perulangan (stop total)
continue	Loncat ke iterasi berikutnya (skip sisa kode)

D. STUDI KASUS

Kasus 1: Faktorial n!

Rumus: $n! = n \times (n-1) \times (n-2) \times \dots \times 1$

Dengan FOR:

```
n = int(input("Masukkan n: "))
hasil = 1
for i in range(1, n + 1):
    hasil *= i
print(f"{n}! = {hasil}")
```

Dengan WHILE:

```
n = int(input("Masukkan n: "))
hasil = 1
i = 1
while i <= n:
    hasil *= i
    i += 1
print(f"{n}! = {hasil}")
```

n	Proses	Hasil
3	$1 \times 2 \times 3$	6
5	$1 \times 2 \times 3 \times 4 \times 5$	120
7	$1 \times \dots \times 7$	5040

Kasus 2: Deret Fibonacci

Rumus: $F_0=0$, $F_1=1$, $F_n = F_{n-1} + F_{n-2}$ Deret: 0, 1, 1, 2, 3, 5, 8, 13, 21, ...

```
n = int(input("Masukkan jumlah suku: "))
a, b = 0, 1
print(a, end=" ")
for i in range(2, n + 1):
    print(b, end=" ")
    a, b = b, a + b
```

Simulasi n=7: | i | a | b | Output | — | — | — | — | — | — | 0 | 1 | 0 | 1 | 1 | 2 | 3 | 1 | 2 | 1 | 4 | 2 | 3 | 2 | 5 | 3 | 5 | 3 | 6 | 5 | 8 | 5 | 7 | 8 | 13 | 8 |

Output: 0 1 1 2 3 5 8

Kasus 3: Cek Bilangan Prima

Bilangan prima hanya habis dibagi 1 dan dirinya sendiri.

```
n = int(input("Masukkan angka: "))

if n < 2:
    print(f"{n} bukan prima")
else:
    prima = True
    for i in range(2, n):
        if n % i == 0:
            prima = False
            break

    if prima:
        print(f"{n} adalah prima")
    else:
        print(f"{n} bukan prima")
```

Uji: | n | Prima? | Alasan | |—|—|—| | 1 | × | < 2 | | 2 | | Hanya habis dibagi 1 dan 2 | | 7 |
| 1×7 saja | | 10 | × | 2×5 = 10 | | 17 | | 1×17 saja |

E. RANGKUMAN

Konsep	Python
FOR 0..n-1	for i in range(n):
FOR 1..n	for i in range(1, n+1):
FOR step	for i in range(awal, akhir, step):
WHILE	while kondisi:
Update counter	i += 1
Break	break — keluar loop
Continue	continue — skip iterasi
Faktorial	hasil *= i
Fibonacci	a, b = b, a+b
Cek prima	FOR i=2..n-1, cek n%i==0

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